Application-aware Slicing for FRMCS: A Deep Reinforcement Learning Approach

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***Abstract*—The Future Railway Mobile Communication System (FRMCS) will replace GSM-R to support safety-critical and high-throughput applications over a limited 5–10 MHz spectrum. Railway services range from ultra-reliable train control, such as European Train Control System and Automatic Train Operation, to bandwidth-intensive video surveillance and best-effort passenger Wi-Fi, each with distinct requirements. Existing network slicing solutions designed for public 5G networks focus on aggregate slice-level guarantees, neglecting heterogeneous application requirements and the strong channel fluctuations induced by high-speed train mobility. To overcome this limitation, we propose in this paper an Application-Driven Slice Scheduling (ADSS) approach tailored for railway communications. ADSS leverages Deep Reinforcement Learning combined with channel-aware resource allocation to dynamically assign Resource Blocks, ensuring application-level Service Level Agreement (SLA) fulfillment. Evaluations on real Signal-to-Noise Ratio traces from trains traveling at speeds up to 350 km/h, demonstrate that ADSS achieves superior application-level SLA satisfaction, reduces violation gaps, and improves spectral efficiency compared to heuristic and state-of-the-art schedulers.**

***Index Terms*—Network Slicing, FRMCS, 5G Railway Networks, Deep Reinforcement Learning (DRL), Resource allocation, QoS assurance.**

I. Introduction

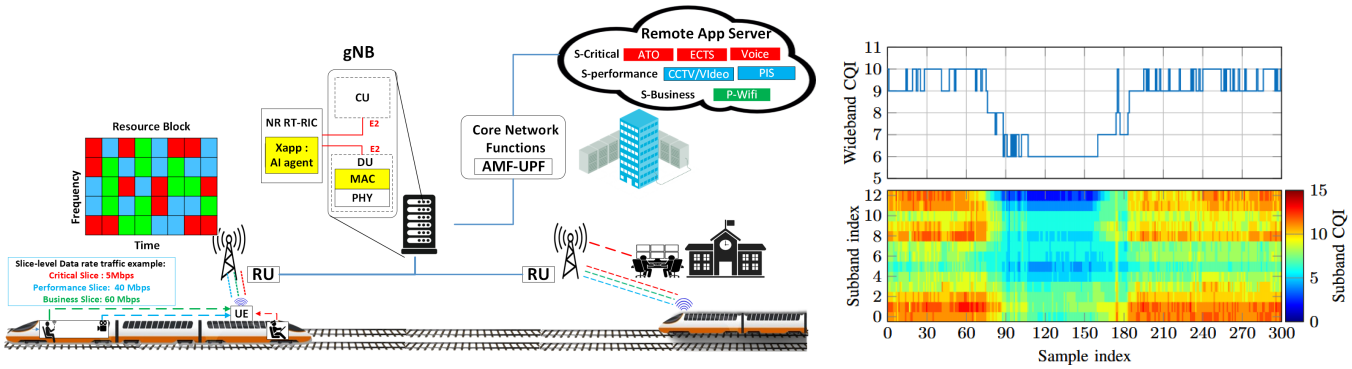
On a train, the communication network linking on-board equipment to the ground infrastructure is known as the train-to-infrastructure network. This network currently relies on the Global System for Mobile Communications – Railway (GSM-R), a 2G-based technology that has been used globally in railway communications for decades and will be phased out by 2035 [1] due to discontinued industrial support [2] and its inability to accommodate the high data rates required by emerging train applications [3]. In response to these limitations, the International Union of Railways (UIC) has proposed the Future Railway Mobile Communication System (FRMCS), a 5G-based solution designed to meet the rigorous requirements of next-generation intelligent and automated railway systems [4].

Railway applications under FRMCS span a wide spectrum, from ultra-reliable signaling such as the European Train Control System (ETCS) and Automatic Train Operation (ATO) to high-bit-rate services like closed-circuit television (CCTV), commonly known as video surveillance, as well as best-effort traffic such as onboard Wi-Fi, each with distinct throughput and latency requirements that must be satisfied simultaneously.

In legacy GSM-R, services were managed separately on the railway's private infrastructure, each with dedicated throughput and latency profiles [5]. With FRMCS, the UIC envisions a unified, 5G-based architecture. A Telecom On-Board Architecture (TOBA) gateway aggregates both safety-critical and high-throughput application traffic and communicates with trackside gNBs within a spectrum as limited as 5–10 MHz [4] (cf. Fig. 1a). Efficiently multiplexing these heterogeneous traffic types over scarce spectrum while maintaining reliable communication at train speeds up to 350 km/h presents unique challenges not typically addressed by conventional public mobile networks. Furthermore, the Radio Access Network (RAN) plays a crucial role since the last-mile wireless link is often the bottleneck in mobile applications [6].

To overcome these challenges, FRMCS can leverage network slicing, a 5G technology that allows a single physical RAN to be partitioned into multiple isolated logical networks, or slices, each optimized for a specific class of services. According to the 3GPP FRMCS TR 22.889 [7], three principal slices could be implemented in railway environments: i) a critical slice providing ultra-reliable, low-latency communications for ETCS, ATO, and emergency voice services; ii) a performance slice, accommodating medium to high bit-rate data streams such as CCTV and Passenger Information Systems (PIS); and iii) a business slice, supporting less time-sensitive services, such as onboard Wi-Fi. By assigning dedicated resource blocks and control-plane functions to each slice, the operator can in principle satisfy SLAs even under severe SNR fluctuations at high train speeds.

Recent works [8]–[10] demonstrate that slice-level guarantees alone do not ensure deterministic application-level SLA satisfaction, which is crucial in railway systems. Despite the maturity of RAN slicing in public 5G networks, existing approaches fall short in the FRMCS context. Application-level frameworks [8], [9] enforce per-flow rate targets but assume 100 MHz carriers and uniform SLAs within a slice, making them ill-suited for 5–10 MHz bands. Popular Channel-aware algorithms such as Best-Channel Quality Indicator (CQI) [11] and proportional-fairness [12] improve spectral efficiency but ignore slice priorities, potentially starving critical traffic. Priority frameworks like NVS [13] enforce hierarchical quotas



(a) Deployment scenario overview. (b) Subband CQI variations.

Fig. 1: Train-to-Infrastructure Network Slicing Use Case.

but rely on static partitions, failing to adapt to rapid CQI variations. Essentially, none of these methods have been evaluated on real high-speed railway subband CQI traces, leaving their performance under extreme mobility unverified.

To address this gap, we propose in this paper an Application-Driven Slice Scheduling (ADSS) framework specifically designed to satisfy heterogeneous application-level demands in railway networks. ADSS leverages Deep Reinforcement Learning (DRL), which is increasingly being adopted in RAN slicing as a replacement for heuristic approaches [14]. Unlike prior works, ADSS explicitly integrates application-level SLA constraints, subband channel awareness, and slice prioritization. Our contributions are threefold:

- We formulate the FRMCS resource allocation and scheduling problem as an Integer Linear Program (ILP) that captures heterogeneous SLAs and slice priorities.
- We design a channel-aware, application-level SLA scheduling algorithm based on DRL with action masking, enabling efficient approximation of the ILP solution and ensuring spectral efficiency under physical constraints.
- We benchmark ADSS against existing approaches using real SNR traces [15] from trains operating at up to 350 km/h, and demonstrate significant improvements in SLA satisfaction, violation reduction, and spectral efficiency.

The remainder of this paper is organized as follows. Section II presents background concepts related to railway networks and application-level RAN scheduling. In Section III, we review existing work relevant to our problem. Our resource allocation problem is formulated in Section IV. Section V explains our reinforcement learning design and discusses its challenges. The performance evaluation of our model compared to existing alternatives is presented in Section VI. Finally, Section VII concludes this paper.

II. Background

A. The Need for Sub-channel-Aware Scheduling

A resource block (RB) is the smallest unit of spectrum that a base station (gNB) can allocate to a user equipment (UE), typically defined over a 1 ms Transmission Time Interval (TTI). The achievable throughput of a UE depends not only on the number of allocated RBs but also on the channel quality on

	Performance slice		Critical Slice	
Train	TOBA 1	TOBA 2	TOBA 3	TOBA 4
Apps	App1	App2	App3	App4
RB1	5	5	10	10
RB2	5	5	10	10
RB3	10	10	5	5
RB4	10	10	5	5

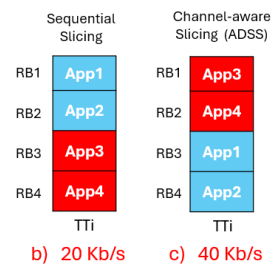


Fig. 2: Conceptual Illustration of the Benefits of Subband-Aware Scheduling.

each RB. In OFDM systems, the channel quality is quantified by the Signal-to-Noise Ratio (SNR) which is mapped to a Channel Quality Indicator (CQI) [16]. The CQI determines the Modulation and Coding Scheme (MCS) selected by the gNB. While the wideband CQI reflects the overall link condition, the subband CQI characterizes the cahnnel quality on a particular RB. As shown in Figure 1b, the same train may experience a stable wideband CQI but heterogeneous subband CQIs, leading to significant differences in achievable per-RB data rates.

This variability highlights the need for subband-aware scheduling. To provide a clearer illustration, we refer to the conceptual example in Fig. 2, comprising four RBs (RB1–RB4), two network slices (Critical and Performance), and four applications (App1–App4). The left 2D grid shows the channel quality, expressed as the achievable data rate (kb/s) per UE and RB. A first-come, first-served scheduler reaches only 20 kb/s, while a subband-aware scheduler that prioritizes higher-CQI RBs doubles throughput to 40 kb/s. This example illustrates how exploiting subband diversity enhances efficiency, a principle that also applies to realistic multi-train scenarios with varying CQIs and SLAs, as we will show in Section VI.

B. Application-level Slicing for Railways

Trains are equipped with TOBA [17], which acts as UEs by aggregating traffic from multiple onboard applications and interfacing with trackside infrastructure. Implementing RAN

TABLE I: Representative FRMCS Application Categories and Requirements.

Slice	Applications	Latency	Throughput
Critical	ETCS, ATO, Emergency Voice, Automatic Train Protection (ATP)	<50 ms	Low (<100 kb/s)
Performance	CCTV, PIS	100–500 ms	Medium–High (Mbps)
Business	Passenger Internet, Entertainment	>500 ms	High (tens of Mbps)

TABLE II: Comparison of Existing RAN Slicing Solutions.

Ref	App-level (a)	Hetero. App-level (b)	Priority (c)	Subband Aware (d)	Railway Mobility (e)	ML (f)
[8], [9]	✓	–	✓	–	–	–
[11], [12]	–	–	–	✓	–	–
[13], [21]	–	–	✓	✓	–	–
[22]	–	–	–	–	–	✓
[23], [24]	–	–	✓	–	–	✓
Our heuristic[10]	✓	✓	✓	✓	✓	–
Our ADSS	✓	✓	✓	✓	✓	✓

slicing in railways therefore requires ensuring SLA compliance at the application level through TOBA gateways. According to the FRMCS specifications in [7], [18], railway services fall into three categories presented in table I: Critical, Performance and Business.

Unlike GSM-R, where applications were supported on separate infrastructures, FRMCS integrates all services into a unified system. Notably, applications within the same slice may have heterogeneous SLAs (e.g., CCTV vs. PIS within the Performance slice). As discussed in [8], [19], many scheduling approaches focus on slice-level guarantees, which cannot ensure deterministic satisfaction at the applications-level. This intra-slice heterogeneity motivates the need for refined, application-aware resource allocation strategies in railway environments.

III. Related works

In this section, we review the literature on RAN slicing along four key dimensions: (i) application-level slicing, (ii) channel-aware resource scheduling, (iii) prioritised slice-level assurance and (iv) the integration of Machine Learning (ML) techniques. To assess their suitability for railway networks, we identify five main criteria, as shown in Table II: (a) application-level approach, (b) heterogeneous throughput requirements within a slice, (c) slice-level priority enforcement, (d) subband-level channel awareness, and (e) explicit evaluation of railway mobility. Due to the scarcity of slicing works in the railway domain, we also compare solutions developed for public 5G networks.

A. Slicing with Application-level assurance

Recent studies have highlighted the importance of application-level guarantees. For instance, Zipper [8] grants each application its own SLA by predicting resource availability with Deep Neural Networks (DNNs) and applying Model Predictive Control (MPC) for RB allocation. Similarly, authors in [9] addressed augmented reality application requirements with a resource manager that adapts bandwidth and edge GPU frequency via a Multi-Armed Bandit algorithm. However, both assume uniform SLA requirements across applications within a slice, which fails to capture the heterogeneity (criteria b) observed in railway systems. Additionally, these solutions target public MNOs with wide bandwidths (e.g., 100MHz), unlike the narrowband (5–10MHz) available in FRMCS. In this narrowband context, subchannel-aware scheduling (criteria d) becomes critical. Therefore, existing application-level frameworks capture application-level requirements but remain unsuitable for railway environments due to their bandwidth assumptions and lack of intra-slice heterogeneity support.

B. Channel-Aware Resource Scheduling

Channel-aware scheduling has long been studied in wireless systems. In many cases, the scheduling problem is NP-hard and a greedy heuristic is adopted, whereby RBs are allocated to UEs that score highest on some given metric. Early works such as [11] use a Best CQI scheduler that maximize throughput by allocating each RB to the UE with the best channel quality, but this can starve UEs in poor channel conditions and do not account for slice priorities (criteria c). Proportional fairness approaches [12] offer more balanced UE allocation using historical throughput to avoid starvation of low-CQI UE but remain UE-centric, overlook application-level SLAs and slice priority (criteria a, b, and c). Beyond these, authors in [20] propose MOHS, a subband-aware hybrid that randomly alternates per-TTI between Round-Robin and Best-CQI with a fixed mixing factor γ. While it improves spectral efficiency over Round Robin and proportional fairness and is fairer than pure Best-CQI, it remains UE-centric and provides no mechanism for application-level SLAs or slice-priority enforcement. Moreover γ is not adapted to traffic/QoS state, and the evaluation does not address railway mobility or narrowband FRMCS constraints. Thus, while channel-aware schemes improve spectral efficiency, they do not address application-level or slice-priority requirements essential for FRMCS.

C. Prioritized Slice-Level Assurance

Priority-based mechanisms such as NVS [13] enforce hierarchical quotas via weighted round-robin scheduling. It assumes that initial slice quotas are defined by the operator as SLA. During resource allocation, any unused quota from a slice can be reallocated to the next slice in the priority hierarchy. However, allocating RB quotas at the slice level without accounting for the heterogeneity of application-level throughput requirements does not guarantee deterministic application-level satisfaction, which is crucial for railway use cases (criteria a). This limitation make NVS inadequate for railways. Extensions such as RadioSaber in [21] combine slice-level quota estimation with subband-aware allocation. However, quota assignment typically scales with traffic demand, inadvertently favoring bandwidth-hungry slices while under-provisioning critical low-rate but high-priority services. In our earlier work [10], we proposed a heuristic tailored to railway environments, which first assigns RBs to the critical slice before allocating remaining resources to performance and business slices. This approach ensured deterministic protection for critical applications and explicitly accounted for railway mobility, but its greedy design resulted in high execution

time and limited scalability (criteria f). Consequently, static or demand-driven quota schemes cannot ensure deterministic protection for critical applications in railway networks (criteria c).

D. Reinforcement Learning-based approaches

The complexity of network slicing has led to growing adoption of Reinforcement Learning (RL) [25] and Deep RL (DRL) [26] . For example, MUESLI [22] formulates scheduling as a Partially Observable Markov Decision Process (POMDP) with multi-objective rewards, balancing SLA, fairness, and spectrum usage. Deep Q-learning approaches [23] introduce slice prioritization, while OnSlicing [24] integrates CQI–MCS mappings for DRL-based orchestration. Yet, these methods either prioritize fairness over criticality (criteria c), lack subband granularity (criteria d), or focus on slice-level rather than application-level assurance (criteria a). Importantly, none have been validated under real high-speed railway channel conditions.

E. Summary

Table II highlights that no existing proposal jointly satisfies all identified requirements (criteria a–f). In particular, application-level heterogeneity (criteria b) and real railway mobility evaluation (criteria e) remain unaddressed in prior works. Our ADSS framework is the first to integrate all these dimensions within a RL–based approach. By training offline and inferring in real time, ADSS achieves lower execution time than our heuristic baseline while outperforming existing state-of-the-art schedulers. Moreover, ADSS is validated with real SNR traces from trains operating at 350 km/h [15], demonstrating its robustness in extreme mobility scenarios, as we will show in Section VI.

IV. Context and Problem formulation

We consider three slices in the FRMCS system: Critical ($s = c$), Performance ($s = p$), and Business ($s = b$). Each slice $s \in \{c, p, b\}$ contains a set of applications A_s with distinct throughput requirements. A train is represented by its TOBA gateway, which aggregates traffic from multiple applications. Resource allocation is performed over K RBs during an allocation horizon of T time slots.

A. Decision Variables

Let the binary decision variable $x_{k,t,i,a,s}$ be defined as:

$$x_{k,t,i,a,s} = \begin{cases} 1, & \text{if RB } k \text{ is allocated at time slot } t \text{ to} \\ & \text{application } a \text{ of train } i \text{ in slice } s, \\ 0, & \text{otherwise.} \end{cases}$$

The achievable throughput for application a of train i on RB k during time slot t is denoted by $\gamma_{i,a}^{k,t}$ (in bits/ms), and is computed according to the 3GPP Standard TS 38.306:

$$\gamma_{i,a}^{k,t} = 10^{-3} \times J \times v_{\text{Llayers}} \times Q_m \times f \times R_{\max} \times \frac{12}{T_s} \times (1 - OH), \tag{1}$$

where J is the number of component carriers, v_{Llayers} represents the maximum number of MIMO layers, Q_m is the modulation order (determined by the subband CQI as in [16]), f is a scaling factor, $R_{\max}$ is the coding rate corresponding to the Train's subband CQI, T_s is the OFDM symbol duration, and OH is the overhead fraction. These parameters are set according to the FRMCS context and drawn from traces as inputs to this equation (1), as explained in our simulations.[1]

B. Related Optimization Problem

Formally, our resource allocation and scheduling problem can be formulated as an Integer Linear Program (ILP) as follows:

$$\min_{x,\{f_{i,a}^s\}} \sum_{k,t,i,a,s} w_s \times x_{k,t,i,a,s} + \sum_{i,a,s} \frac{1}{w_s} f_{i,a}^s \tag{2}$$

$$\text{s.t:} \sum_{s,a} x_{k,t,i,a,s} \leq 1, \quad \forall k, t, i \tag{3}$$

$$\sum_{k,t} x_{k,t,i,a,s} \times \gamma_{i,a}^{k,t} + f_{i,a}^s \geq \text{SLA}(a, i, s) \quad \forall a \in A_s \tag{4}$$

$$f_{i,a}^s = \frac{\left|\min\left(0, \sum_{k,t} x_{k,t,i,a,s} \times \gamma_{i,a}^{k,t} - \text{SLA}(a, i, s)\right)\right|}{\text{SLA}(a, i, s)} \tag{5}$$

$$x_{k,t,i,a,s} \in \{0, 1\} \qquad \forall k, t, i, a, s \tag{6}$$

The objective function in (2) contains two terms. The first term minimizes the total weighted resource usage of the three slice types (i.e., the cost of resource consumption) using the weight set $W = \{w_c, w_p, w_b\}$. This set is defined such that $w_c < w_p < w_b$ implying lower cost in minimization of objective for critical slice, with values ranging between 0 and 1 and having a sum equal to 1 ($w_c + w_p + w_b = 1$). This is enforcing a priority order where the Critical slice has higher precedence than the Performance slice, which in turn is prioritized over the Business slice. The second term represents a penalty component, where $f_{i,a}^s \geq 0$ denotes the throughput penalty for application a of train i in slice s.

Constraint (3) ensures that each RB can be allocated to at most one application in a time slot t. Constraint (4) ensures that the cumulative throughput for each application a of train i in slice s must meet its SLA demand $\text{SLA}(a, i, s)$. Constraint (5) defines the violation penalty $f_{i,a}^s$, which must be positive. Finally, Constraint (6) indicates that $x_{k,t,i,a,s}$ is a binary variable.

The resulting ILP formulation captures the fundamental trade-off of allocating limited RBs among heterogeneous applications while enforcing slice priorities and minimizing SLA violations. However, its exponential computational complexity ($\mathcal{O}(2^{K \cdot T})$) makes it impractical for real-time deployment in railway environments. This limitation motivates the development of a learning-based approach, presented in Section V.

V. Proposal: ADSS

In this section, we detail our proposed Application-Driven Slice Scheduling (ADSS) framework by first formulating the

[1] Network configuration is given in table III, whereas the 64 QAM mapping table proposed in [16] is used to map train SNR traces to subband CQI.

problem as a Markov Decision Process (MDP) and discussing the associated complexity challenges. This allows the use of Reinforcement Learning (RL) to approximate the ILP solution defined in Section IV while ensuring tractability under real-time railway conditions.

A. MDP Formualation and Complexity Considerations

The scheduling problem can be represented by an MDP $M = (\mathcal{S}, \mathcal{A}, P, R)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ the action space, P the transition probability governed by the railway channel dynamics, modeled through SNR traces (see Section VI), and R the reward. Three challenges arise and must be considered:

- Large Action Space: Allocating K RBs to N applications at each time slot in an application-level scheduler would yield an action space of size $|\mathcal{A}| = O(|A|^{KN})$, which is exponential in K and N. This complicates exploration and increases the computational burden for the RL agent.
- High-Dimensional State: The scheduler must track channel conditions, per-application SLA progress, and achieved data rates. Storing these metrics, along with the estimated data rates $\gamma_{i,a}^{k,t}$ for each application, results in a high-dimensional state space.
- Heterogeneous Requirements: Different applications require distinct resource allocations. Even within the same slice, assuming identical SLAs for all applications is an oversimplification that can lead to SLA violations.

To address these challenges, ADSS integrates action masking, SLA-driven rewards, and offline training. These different components will be described in the next subsections.

B. State Representation

At each time slot t, the system state $s_t \in \mathcal{S}$ includes:

1) Time Index: It corresponds to the current slot within the allocation horizon T (e.g., $T = 10$ ms in our simulations, a typical frame length in 5G).
2) Achievable Data Rates: $\gamma_{i,a}^{k}$ for each RB k, application a, and train i, are derived from CQI values (cf. Equation 1).These values are computed based on the train channel conditions via periodic or aperiodic feedback as specified in 3GPP TS 36.213, or they can be obtained through historical measurements and predictions as in [27].
3) Allocation Progress: For each active application a of train i in slice s, we maintain cumulative throughput $A(i, a)$ already achieved by each application. This metric indicates how closely the application meets its SLA within the allocation episode.
4) Mask Flags: We introduce a binary flag for each application indicating whether it still requires additional resources or not. Once an application's SLA is met at time step t, its corresponding mask is set to 0, and it is excluded from subsequent allocations in the current episode.

These observations are scaled and structured as vectors of length K (the number of RBs). The agent's policy network processes these vectors via fully connected and multi-headed layers to produce both the allocation action and a state value estimate. As discussed in survey [28], there is a consensus in the literature to include the demand metrics in the MDP state in RL-based proposal.

C. Action Space with Masking

At each time step, the agent must decide which application receives a RB k. Without pruning invalid actions, the naive action space grows exponentially (cf. Section V-A). To reduce complexity, ADSS employs action masking which is described in [29]. This technique filters out invalid infeasible actions (e.g., assigning RBs to already satisfied applications). In practice, we pre-select K candidate applications (one for each RB) for each slice (Critical, Performance, and Business). This yields a candidate matrix $A_{\text{candidates}}$ of size $K \times C$ (with $C = \{0, 1, 2, 3\}$; here, 0 indicates no allocation, and 1, 2, 3 correspond to the Critical, Performance, and Business slices, respectively). Thus, the overall action space is reduced to a manageable discrete size of $K \times 4$.

Consequently, the agent's action $\mathbf{a}_t$ at time t is an integer vector of length K with each element chosen from the set of candidate actions in $A_{\text{candidates}}$. For example, the action output for each RB in K might be:

$$a_t \in \{0,\, a_{1,1},\, a_{2,3},\, 0,\, a_{3,s},\, \ldots,\, a(k-2, c),\, a(k-1, c)\},$$

where the environment maps the integer (e.g., `0` indicating no allocation on the given RB, or `1`, `2`, `3` corresponding to allocations to the Critical, Performance, and Business slices, respectively). The model's output is an array of K allocations corresponding to the number of RBs allocated at time step t.

Moreover, the $A_{\text{candidates}}$ matrix is updated at each time slot based on current channel conditions, ensuring that the best candidate application for each RB is selected to maximize spectral efficiency. This significantly reduces exploration complexity and accelerates convergence.

D. Reward Function

The reward guides the agent toward satisfying application-level SLAs while avoiding resource waste. Let $\Delta(a, i, s)$ denote the SLA satisfaction ratio:

$$\Delta(a, i, s) = \frac{D(a, i, s)}{\text{SLA}(a, i, s)} \tag{7}$$

where $D(a, i, s)$ is the achieved throughput so far by application a of train i in slice s, and $\text{SLA}(a, i, s)$ is its data rate requirement.

For each slice s, a priority weight ω_s is defined with $\omega_c > \omega_p > \omega_b$. The reward at time t is:

$$R(t) = \sum_{a,i,s \in \mathcal{A}} \begin{cases} \omega_s \cdot \dfrac{1}{\Delta(a, i, s)} & \text{if } \Delta(a, i, s) \geq 1 \\ -\omega_s \cdot \big(1 - \Delta(a, i, s)\big) & \text{if } \Delta(a, i, s) < 1 \end{cases} \tag{8}$$

In this formulation, if $\Delta(a, i, s) \geq 1$, the agent receives a small positive reward that decreases with increasing $\Delta(a, i, s)$. This choice ensures diminishing returns when SLAs are exceeded, discouraging resource over-provisioning. Conversely,

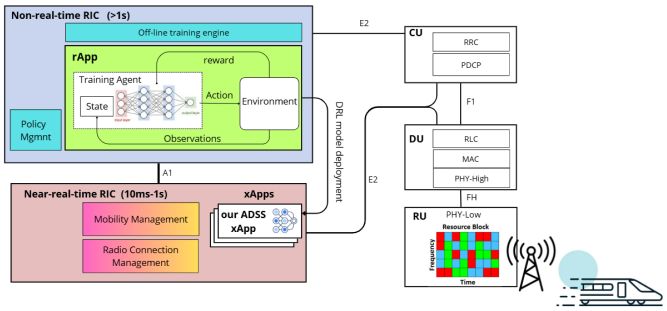


Fig. 3: Potential ADSS Deployment Scenario.

if $\Delta(a, i, s) < 1$, a negative reward is incurred proportional to the SLA shortfall.

Overall, the proposed reward function prioritizes the critical and performance slices by assigning them higher weights, while allowing a best-effort approach for the business slice, thereby guiding the agent towards efficient and balanced resource allocation.

E. Deployment Considerations

The proposed ADSS scheduler targets deployment at the gNB edge along railway tracks, as illustrated in Fig. 1a. With 5G and beyond, railway networks are expected to adopt the O-RAN architecture, in which a Centralized Unit (CU) connects to multiple Distributed Units (DUs), each managing several Radio Units (RUs). We take a closer look at the deployment environment, illustrating the interfaces between RAN components in Fig. 3. In this environment, the Near-Real-Time RIC hosts xApps that provide closed-loop RAN control, enabling local optimization under high-mobility conditions.

In this context, ADSS can be implemented as an xApp operating at the near-real-time RIC, as shown in Fig. 3. Its role is to schedule RBs among Critical, Performance, and Business applications while dynamically adapting to local CQI variations observed by each train's TOBA gateway. The xApp interacts with the gNB scheduler to enforce heterogeneous application-level SLA within a few milliseconds, ensuring their satisfaction even under rapidly changing railway channels.

A key advantage of this deployment model is that computationally expensive policy training can be performed offline on collected railway SNR traces [15], while only lightweight inference is executed in real time at the gNB. This separation ensures that the scheduler meets strict latency constraints, avoids service disruption during training, and remains scalable across multiple gNBs along the railway line. Moreover, the distributed nature of O-RAN allows each gNB to adapt locally to its channel conditions, while global SLA priorities are preserved through slice weights. The performance of this deployment model is assessed in Section VI.

VI. Performance Evaluation

This section evaluates ADSS against optimal and state-of-the-art schedulers using real railway channel traces. We first describe the baselines and simulation setup, define the metrics, outline the training methodology used in our comparisons, and then present the simulation results.

A. Baselines

We benchmark two different implementations of ADSS against five state-of-the-art approaches that are described in Section III:

1) Optimal: The formulated ILP problem is solved using Gurobi with the well known branch-and-bound algorithm. To prevent excessive computation time, the solver runtime is limited to 10 s per allocation instance.
2) ADSS-PPO: This ADSS version uses Proximal Policy Optimization [30] for model training.
3) ADSS-DQN: This ADSS version uses Deep Q-Network [31] for model training.
4) HSRS [10]: This scheme corresponds to our previously proposed heuristic designed for FRMCS, which first minimizes SLA slack for Critical slice and then maximizes the weighted throughput for the remaining slices.
5) RadioSaber [21]: This scheme reflects a state-of-the-art channel-aware slice scheduling approach.
6) Best-CQI [11]: This scheme corresponds to a greedy scheduler assigning each RB to the highest-CQI application.
7) NVS [13]: This scheme enforces priority quotas through a weighted round-robin mechanism.

It is worth noting that the Proportional Fairness approach [12] is omitted because fairness may compromise critical SLAs, which is unacceptable in railway environments. These baselines collectively cover optimal, heuristic, channel-aware, quota-based, and RL-driven strategies, providing a comprehensive comparison.

B. Simulation Environment

We simulate a 5 MHz bandwidth to realistically model limited spectrums allocated to railways network operator in Europe [32] in the 900 MHz band. We configured our simulation according to the parameters listed in Table III. In our simulations, connected applications and gNB allocation decisions are evaluated on a per-frame basis, with each frame lasting 10 ms in accordance with 5G standards. Although railways application SLAs are typically expressed in Kbps for end-to-end communications, we convert these requirements to their per-frame equivalents for RAN-level measurements, as shown in Table IV.

Regarding implementation, we developed a test environment on a Laboratory system equipped with 36 GB of RAM and a 64 core CPU, using Python. The code is made available in [33] to support reproducibility and reuse within the research community. We developed a subband CQI loader compliant with 5G standards based on real SNR dataset[15]. These SNR data are loaded by the subband CQI loader and converted to subband CQI using SNR-CQI-MCS mapping from [16]. As specified in 3GPP TS 36.213, subband CQI can be reported to gnB periodically. To avoid control plane overhead, subband CQI is usually computed not for a single RB but for a Resource

TABLE III: Simulation Parameters.

Parameter	Value	Description
Frequency bandwidth	5 MHz	Band allocated in the 900 MHz band
K	25	Number of Resource Blocks (RBs)
T	10	Number of time steps per simulation episode
J	1	Number of component carriers
v_{Llayers}	1	Maximum number of MIMO layers
μ	0	Numerology (0 corresponds to 15 kHz subcarrier spacing)

TABLE IV: Service Level Agreements Expected for Different Applications.

Slice	Applications	Data Rate (Bits/frame)
Critical	ETCS	400
	Voice	900
Performance	CCTV	10 000
	PIS	20
Business	Passenger WiFi (PWiFi)	20 000

Block group (RBG) which can be a group of 4, 6 or 8 RBs. Following this periodic CQI feedback in our environment, at each configurable time index (which is set by default to 10ms), every TOBA gateway in the simulation is assigned subband CQIs extracted from the real dataset of a specific Train CSV file. This subband CQI loader enables the computation of achievable throughput per RB, thereby facilitating channel-aware scheduling.

Regarding the traffic scenario, a block section in railway operations is the safety segment of the railroad where only one train is allowed on the track, which is typically 1.5–2 km long under dense traffic conditions to maintain safe separation between trains. Since a gNB typically covers 10–15 km in rural areas [34] (and less in urban areas), a distributed gNB must support approximately ten train block sections. Accordingly, as summarized in Table V, we consider a dense scenario with 15 trains under a single gNB, each generating traffic for multiple railway applications (26 in total). Under average channel conditions (CQI = 7), the cumulative throughput demand (SLA) of these applications cannot be fully satisfied by the available 5 Mhz bandwidth (which corresponds to a total of 25 available RBs in the system). With channel conditions updated every frame based on the CQI, this setup creates a dynamic, overloaded scenario, enabling clear observation of resource contention and potential SLA violations.

C. Evaluation Metrics

We adopt four metrics to assess service satisfaction and spectrum efficiency. Considering that $r_{a_{i,s}}$ denote the achieved throughput of application a of train i in slice s, and $d_{a_{i,s}}$ its SLA demand, they can be defined as:

- SLA Violation Rate (VR):

$$\text{VR} = \mathbb{E}\big[\mathbf{1}\{r_{a_{i,s}} < d_{a_{i,s}}\}\big],$$

  It corresponds to the probability that an application fails to meet its SLA.
- Violation Gap (VG):

$$\text{VG} = \mathbb{E}\big[(d_{a_{i,s}} - r_{a_{i,s}})_+\big], \quad (x)_+ = \max(x, 0),$$

  It measures the average shortfall during violations, thereby demonstrating the severity of the occurred violations.

TABLE V: Simulation scenario

Scenario	TOBA per Slices	Nbr of Apps across TOBAs	Total Apps
15 Trains (1 train = 1 TOBA gateway)	Critical: 6	6 ETCS + 6 Voice	26 apps
	Performance: 5	5 CCTV + 5 PIS	
	Business: 4	4 PWiFi	

TABLE VI: Computational Complexity and Average Execution Time per Allocation Slot.

Algorithm	Complexity	Avg Exec. Time (ms)
Optimal	$O(2^{NKT})$	694.4
ADSS_PPO	$O(TKN)$	14
BestCQI	$O(TKN)$	3.4
HSRS	$O(TKN)$	3.0
NVS	$O(TKN)$	2.4
RadioSaber	$O(TKN)$	1.5
ADSS_DQN	$O(TKN)$	1.2

- Average Throughput:

$$\overline{r}_{a_{i,s}} = \mathbb{E}[r_{a_{i,s}}],$$

  It corresponds to the mean rate per application over time, showing long-term fairness of RBs allocations.
- Spectral Efficiency (SE):

$$\text{SE}_s = \frac{\sum_{i,a\in s} r_{a_{i,s}}}{\sum_{a\in s} \text{RB}_a},$$

  It corresponds to the average bits per RB achieved by slice s in train i, reflecting efficiency under scarce spectrum.

It is worth noting that while VR and VG capture reliability, the average throughput metric emphasizes fairness across applications, and the SE metric measures spectrum efficiency. Together, these performance metrics comprehensively reflect the requirements of FRMCS.

D. Simulation Results

1) Complexity and Runtime Analysis: Algorithmic scalability and inference latency are critical in near-real-time RAN control. Table VI reports both asymptotic complexity and measured execution time. We can see that the optimal approach grows exponentially with the number of RBs and applications, requiring on average 694 ms per slot, which is infeasible for online deployment. All other algorithms, including ADSS, scale linearly in $O(TKN)$. Runtime measurements confirm that ADSS-DQN executes in 1.2 ms per slot, well within the 1–10 ms budget of near-real-time RIC functions in O-RAN, while ADSS-PPO incurs higher inference cost (14 ms). These results demonstrate that ADSS achieves the same complexity order as heuristics while maintaining runtime feasibility in railway environments.

2) Training Analysis: We begin by analyzing the training behavior of ADSS using DQN and PPO. Fig. 4 presents the episodic reward evolution. We can see that PPO demonstrates faster convergence and earlier stabilization (around 45 vs.

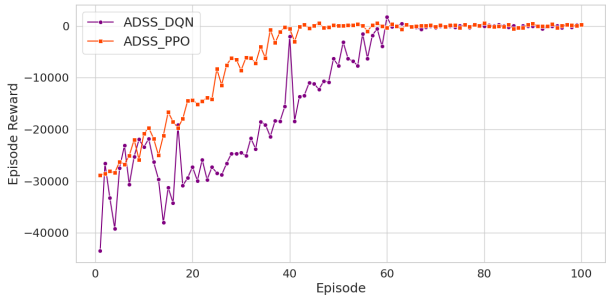


Fig. 4: Episodic Training Rewards: ADSS-DQN vs ADSS-PPO. PPO converges faster but incurs higher inference cost.

TABLE VII: Training Hyperparameters

Common hyperparameters			
Parameter	**Value**	**Parameter**	**Value**
Random seed	42	Optimizer	Adam
Learning rate	0.005	Discount γ	0.99
Mini-batch size	64		

Specific hyperparameters		
Parameter	**PPO**	**DQN**
Network	Actor: 8-128-128-4; Critic: 8-128-128-1; ReLU	Q: 8-128-128-4; ReLU
Policy/value	clip ϵ=0.2; GAE λ=0.25; c_{ent}=0.02; c_V=0.5	1-step TD; soft target τ=0.005
Replay buffer	–	50,000
Exploration	–	ϵ: 1.0→0.05 (10k)

62 episodes), whereas DQN shows higher variance due to replay buffer mixing under non-stationary channel conditions. The clipped objective and on-policy updates in PPO enhance training stability but come at the cost of higher inference complexity due to its actor–critic architecture. In contrast, DQN employs a single Q-network, resulting in lower inference overhead despite more fluctuating training performance.

Table VII summarizes the hyperparameters used for reproducibility. Both PPO and DQN employ comparable neural network architectures to ensure a fair comparison. As training is performed offline while inference remains time-critical, ADSS-DQN provides the optimal deployment trade-off offering sufficiently stable training with minimal runtime overhead. Consequently, ADSS-DQN is adopted in subsequent evaluations as the default ADSS implementation. In the following, we discuss the performance of all schedulers with respect to the four metrics defined in Section VI-C.

3) SLA Violation Rate: Fig. 5 illustrates the violation rates across slices. We can see that the Best-CQI scheduler strongly favors RBs with high channel conditions, leading to starvation of applications with weaker CQIs and consequently high violation rates up to 68% in the performance slice. NVS enhances reliability for critical services by enforcing fixed quotas (50% critical, 30% performance, and 20% business), yet still experiences 50% violations in the performance slice due to their high throughput demands. RadioSaber, though channel-aware, is priority-agnostic and allocates resources proportionally to traffic demand, resulting in 25% violations

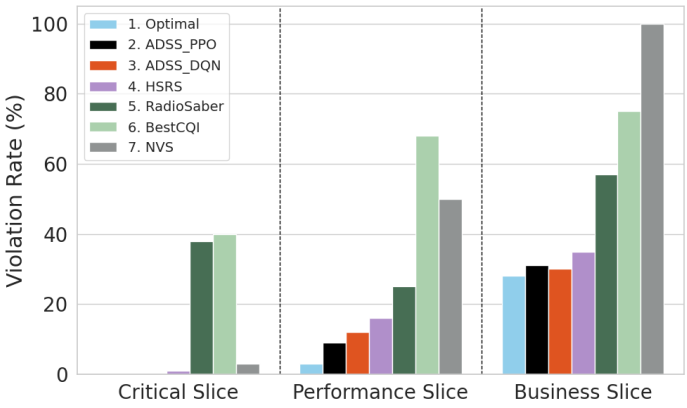


Fig. 5: SLA Violation Rate

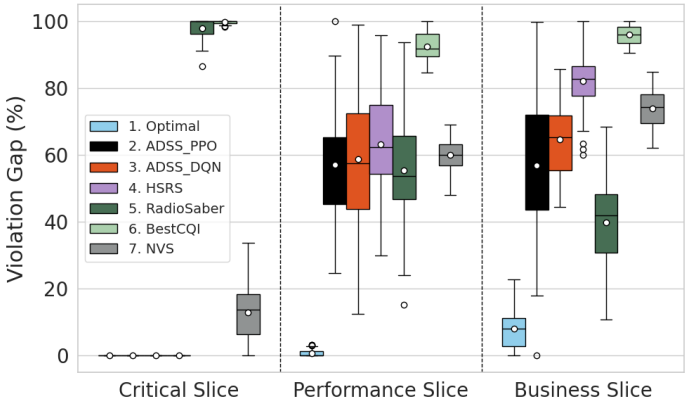


Fig. 6: Violation Gap

in the performance slice and occasional violations even within critical traffic. On the other hand, HSRS ensures full satisfaction of critical traffic but still incurs 16% violations in the performance slice. In contrast, ADSS-DQN achieves only 12% violations in the performance slice, representing reductions of 25% compared to HSRS, 52% compared to RadioSaber, 76% compared to NVS, and 82% compared to Best-CQI. For the business slice, ADSS-DQN records 30% violations, closely matching the optimal approach of 28%. The violations observed in the optimal baseline confirm that these errors stem from inherent overload within the 5 MHz band rather than limitations of the scheduling algorithms.

4) Violation Gap: Fig. 6 quantifies the severity of SLA violations. We can see that the Best-CQI scheduler exhibits the largest gaps, since applications with poor CQI often receive no resources, resulting in deficits of several kilobits per frame. NVS reduces the gap in the critical slice thanks to its fixed quotas, but this comes at the expense of performance and business slices, where large shortfalls remain. RadioSaber, being channel-aware but priority-agnostic, produces severe gaps in the critical slice, since resources are steered towards bandwidth-hungry services. On the other hand, HSRS eliminates the gap in the critical slice but still incurs deficits in performance and business traffic. In contrast, ADSS eliminates all gaps in the critical slice and reduces the

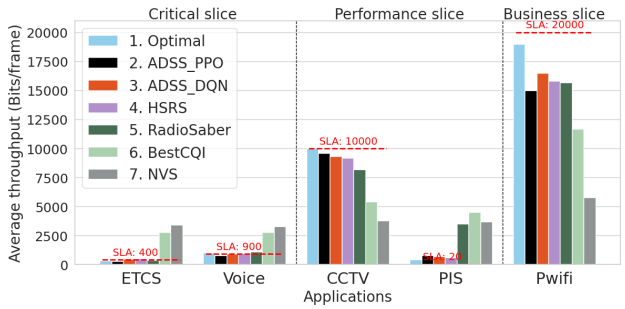


Fig. 7: Average Throughput per Application

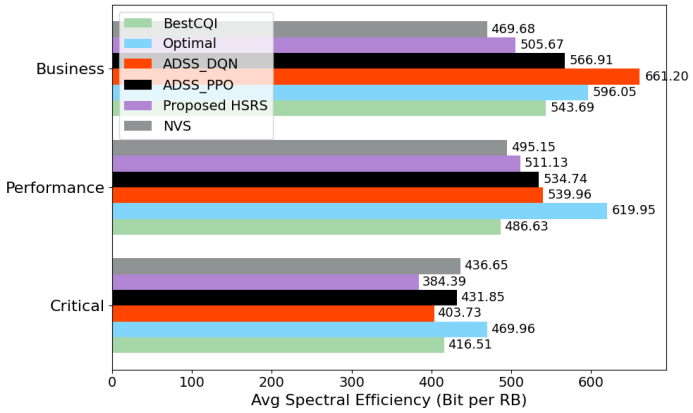


Fig. 8: Spectral Efficiency per Slice

average shortfall in performance and business slices to nearly half of that observed under HSRS. This confirms that ADSS not only lowers violation frequency (Fig. 5) but also minimizes the severity of unmet demand when overload occurs.

5) Average Throughput per Application: Fig. 7 reports the average throughput achieved per application. We can observe that both Best-CQI and NVS consistently underserve high-demand performance applications such as CCTV, which on average receive only about half of their SLA. However, HSRS improves throughput for critical traffic but sacrifices performance and business slices, resulting in under-provisioning of applications like PIS and WiFi. On the other hand, ADSS achieves a more balanced distribution. Indeed, critical applications consistently reach their targets, while performance applications (e.g., CCTV) achieve on average 90% of their SLA, compared to only 65% under NVS and 50% under Best-CQI. Business traffic is also served more fairly. In fact, WiFi throughput under ADSS is 15–20% higher than with HSRS. These results demonstrate that ADSS effectively aligns long-term resource allocation with both application-level SLAs and slice priorities.

6) Spectral Efficiency: Finally, Fig. 8 evaluates spectral efficiency per slice, expressed in bits per RB. In the critical slice, ADSS-DQN achieves 403.7 bits/RB, which corresponds to approximately 86% of the optimal value (469.9 bits/RB) and about 5% higher than HSRS (384.4 bits/RB). In the performance slice, ADSS-DQN reaches 540 bits/RB, outperforming HSRS (+6%), RadioSaber (+11%), and NVS (+17%), while achieving 87% of the optimal value (620 bits/RB). Even in the business slice, ADSS performs efficiency compared to all baselines, reaching 580 bits/RB, i.e., about 12% higher than HSRS and 17% higher than NVS. These results demonstrate that our proposal attains near-optimal spectral efficiency in prioritized slices while maintaining high efficiency for best-effort traffic. Such near-optimal performance under tight spectrum constraints is particularly significant for FRMCS, where operators must maximize the resource utilization within the 5–10 MHz bandwidth.

Overall, our proposed ADSS approach consistently outperforms heuristics and state-of-the-art schedulers across all metrics. It ensures zero violations in critical applications, reduces both violation rate and gap in performance and business, balances throughput allocation more effectively, and achieves spectral efficiency close to the optimal upper bound in prioritized slices. Combined with its low inference latency (1.2 ms with DQN), these results demonstrate that ADSS is a practical and effective scheduler for FRMCS deployment.

VII. Conclusion

In this paper, we proposed an Application-Driven Slice Scheduling (ADSS) approach designed for railway networks, where resource constraints and heterogeneous application requirements pose significant challenges. By formulating the scheduling problem as a Markov Decision Process (MDP) and integrating action masking along with an SLA-driven reward function, our approach dynamically allocates resource blocks (RBs) across critical, performance, and business slices. This design effectively addresses the issues of a large action space and high-dimensional state representations, ensuring that the unique data rate demands of each application are met under varying conditions.

Extensive evaluation on real railway channel traces has shown that ADSS fully eliminates violations in critical traffic, substantially reduces violations in performance applications, and achieves near-optimal spectral efficiency, all while satisfying near-real-time runtime constraints. Compared to classical schedulers (Best-CQI, NVS), state-of-the-art methods (RadioSaber), and an heuristic approach (HSRS), ADSS consistently achieves higher SLA satisfaction and efficiency under overloaded railway scenarios.

As future work, we plan to extend ADSS to support MIMO and Coordinated Multi-Point (CoMP) configurations, aiming to further enhance spectral efficiency and robustness at cell edges. Additionally, integrating predictive mechanisms that leverage train signaling information (e.g., ETCS) could enable proactive resource allocation and improved handling of emergency scenarios. These advancements would further establish ADSS as a robust foundation for reliable, application-aware RAN slicing in FRMCS and next-generation 6G-enabled intelligent railway systems.

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